

Praveenkumar Venkatesan

Doctoral Student
Centre of Excellence - Critical
Transitions in Complex Systems
Department of Aerospace Engineering
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EDUCATION

Doctor of Philosophy Indian Institute of Technology Madras, Chennai, India	2024- Present CGPA: 8.5/10
Master of Science in Physics Bharathidasan University, Tiruchirappalli, India	2020-2022 CGPA: 8.22/10
Bachelor of Science in Physics Muthayammal College of Arts and Science, Periyar University, Salem, India	2017-2020 CGPA: 8.64/10

RESEARCH INTERESTS

- Complex Networks
- Climate Processes and Dynamics
- Nonlinear Dynamics and Chaos

PROFESSIONAL EXPERIENCE

Predoctoral researcher Principal Investigator: Prof. R. I. Sujith Centre of Excellence for Critical Transitions in Complex Systems Department of Aerospace Engineering Indian Institute of Technology Madras	July, 2022 - July, 2024
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PUBLICATIONS

- [1] **P. Venkatesan**, G. Chopra, R. Ravindran, S. Gupta, V. R. Unni, N. Marwan, J. Kurths, and R. Sujith, “[The circular movement of synchronous extreme precipitation preceding Kerala floods in 2018 and 2019](#)”, *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 35, no. 5, 2025.
- [2] G. Chopra, V. R. Unni, **P. Venkatesan**, S. M. Vallejo-Bernal, N. Marwan, J. Kurths, and R. Sujith, “[Community structure of tropics emerging from spatio-temporal variations in the Intertropical Convergence Zone dynamics](#)”, *Scientific Reports*, vol. 14, no. 1, p. 24 463, 2024.

CONFERENCE PRESENTATIONS

1. **P. Venkatesan**, G. Chopra, R. Ravindran, S. Gupta, V. R. Unni, N. Marwan, J. Kurths, and R. I. Sujith. “Time-varying functional climate networks capture emergent phenomena prior to the Kerala floods in 2018 and 2019.”, *In Conference on Nonlinear Systems and Dynamics*, Bharathidasan University, Tiruchirappalli, India, March 10–13, 2025

2. **P. Venkatesan**, G. Chopra, R. Ravindran, S. Gupta, V. R. Unni, N. Marwan, J. Kurths, and R. I. Sujith. “Time-varying network captures emerging phenomena prior to the Kerala floods in 2018 and 2019.”, *In NetSci X 2025*, Indian Institute of Technology, Indore, India, January 14–17, 2025
3. M. Mohammed Mujahid, **P. Venkatesan**, G. Chopra, and R. I. Sujith. “Spatiotemporal dynamics of deep convective clouds across a wide range of scales using multiresolution analysis.”, *In NetSci X 2025*, Indian Institute of Technology, Indore, India, January 14–17, 2025
4. Gaurav Chopra, V. R. Unni, **P. Venkateshan**, S. M. Vallejo-Bernal, N. Marwan, Jürgen Kurths, and R. I. Sujith. “Classification of Tropics Based on the Spatio-Temporal Dynamics of the Intertropical Convergence Zone from a Complex Networks Perspective.”, *104th Annual AMS Meeting 2024*, volume 104, page 435138. American Meteorological Society, January 30, 2024

SKILLS

- **Programming language:** FORTRAN, C/C++, Python, Parallel Computing, Matlab and Simulink, GNUPlot, HTML.
- **Time series analysis:** Linear/Non-linear time series analysis.
- **Data handling:** Climate Data, Experimental data acquired from lab-scale gas turbine engines
- **Type setting:** L^AT_EX, MS Word

CONFERENCES AND WORKSHOPS ATTENDED

Bharathidasan University, Tiruchirappalli Workshop on Machine Learning in Physics: Innovations and Applications	12 - 13 December 2024
Indian Institute of Technology Madras (IIT Madras) Perspectives in Nonlinear Dynamics (PNLD) - 2023	01 - 04 August 2023
International Centre for Theoretical Sciences (ICTS - TIFR) Workshop on Mathematical modelling of Climate, Ocean and Atmosphere processes	26 - 30 June 2023

REFEREES

Dr. R. I. Sujith

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D. Srinivasan Institute Chair Professor,*
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